

Effect of handling and processing on pesticide residues in food- a review

Pesticides are one of the major inputs used for increasing agricultural productivity of crops. The pesticide residues, left to variable extent in the food materials after harvesting, are beyond the control of consumer and have deleterious effect on human health. The presence of pesticide residues is a major bottleneck in the international trade of food commodities. The localization of pesticides in foods varies with the nature of pesticide molecule, type and portion of food material and environmental factors. The food crops treated with pesticides invariably contain unpredictable amount of these chemicals, therefore, it becomes imperative to find out some alternatives for decontamination of foods. The washing with water or soaking in solutions of salt and some chemicals e.g. chlorine, chlorine dioxide, hydrogen peroxide, ozone, acetic acid, hydroxy peracetic acid, iprodione and detergents are reported to be highly effective in reducing the level of pesticides. Preparatory steps like peeling, trimming etc. remove the residues from outer portions. Various thermal processing treatments like pasteurization, blanching, boiling, cooking, steaming, canning, scrambling etc. have been found valuable in degradation of various pesticides depending upon the type of pesticide and length of treatment. Preservation techniques like drying or dehydration and concentration increase the pesticide content many folds due to concentration effect. Many other techniques like refining, fermentation and curing have been reported to affect the pesticide level in foods to varied extent. Milling, baking, wine making, malting and brewing resulted in lowering of pesticide residue level in the end products. Post harvest treatments and cold storage have also been found effective. Many of the decontamination techniques bring down the concentration of pesticides below MRL. However, the diminution effect depends upon the initial concentration at the time of harvest, substrate/food and type of pesticide. There is diversified information available in literature on the effect of preparation, processing and subsequent handling and storage of foods on pesticide residues which has been compiled in this article.